

Henry Chen

778-681-4165 | hchen285@my.bcit.ca | <https://www.linkedin.com/in/henry-chen-832a89386/>
<https://github.com/Chenry513>

EDUCATION

Master of Science in Applied Computing (CGPA: 91%) British Columbia Institute of Technology	Sept. 2025 – Dec. 2026 (est.) Burnaby, BC
Bachelor of Science in Computing Science Simon Fraser University	Sept. 2020 – Aug. 2025 Burnaby, BC

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C/C++, SQL, Bash

Web & Frameworks: Django, Django REST Framework, Vue.js, Next.js, Flask, REST APIs

Machine Learning & Data: scikit-learn, pandas, numpy, Gradient Boosting, ETL pipelines, cross-validation, feature engineering

Databases & DevOps: PostgreSQL, MySQL, SQLite, Git, GitHub Actions, CI/CD, Linux, Vercel, Render

Testing & Quality: Google Test (C++), unit testing, TDD, schema validation, code coverage

WORK EXPERIENCE

Mitacs Research Intern – Robotics & Computer Vision BCIT — Seaspan Shipyards Collaboration (Supervisor: Mathew Smith)	Jan. 2026 – Present Vancouver, BC
- Developing a 3D weld joint reconstruction pipeline using VGGT (Meta Research) to extract geometric coordinates for robotic arm path planning. - Built WeldPath Viz, a browser-based 3D point cloud tool for inspecting reconstructed joints and defining weld seam start/end coordinates for automated welding. chenry513.github.io/weldpath-viz/visualize_weld.html - Designed a calibration system to translate reconstructed scene coordinates into real-world millimeter measurements for robotic execution.	
Research Assistant (Backend & Automation Engineer) Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent) dl-tools.onrender.com	May – Aug. 2025 Burnaby, BC
- Developed an automated ingestion pipeline validating structured user submissions against defined requirements prior to production integration, reducing manual review effort by 60%. - Built a Python-based schema validation tool to enforce expected data behavior and prevent invalid content from entering system workflows. - Integrated GitHub Actions CI to execute automated validation checks on each submission, supporting consistent system outcomes. - Maintained and extended an existing Django codebase while preserving system stability and documenting processes for non-technical contributors.	
Research Assistant (Backend Engineer) Simon Fraser University — PadComputing Lab (Prof. Nicholas Vincent)	Aug. – Dec. 2024 Burnaby, BC
- Developed backend components for a Django-based research platform supporting structured LLM metadata and user-facing functionality. - Designed relational models and schema logic to support consistent data behavior and maintainable structured storage. - Implemented 8+ RESTful API endpoints using Django REST Framework, translating functional requirements into reliable system interactions. - Managed migrations and enhanced Django Admin tooling to maintain data integrity and support stable platform evolution.	

PROJECTS

Full-Stack Personal Project

readmeify — AI-Powered README Generator

Feb. 2026

readmeify-five.vercel.app

- Built and deployed a full-stack web app that produces tailored README documentation by reading actual repo structure, dependencies, and file contents — eliminating manual documentation overhead.
- Implemented GitHub OAuth for authentication and the GitHub API to extract file trees and dependency manifests as structured context, enabling accurate and project-specific generation.
- Integrated Groq API (LLaMA 3.3 70B) with real-time streaming output and built with Next.js, NextAuth, and deployed on Vercel with secure server-side environment variable handling.

Data Science & ML Personal Project

CardioScan — Cardiac Risk Assessment Dashboard

Dec. 2025

cardiac-assessment.onrender.com

- Designed an ETL pipeline extracting and combining 4 UCI data sources, transforming inconsistent encodings across institutions, and imputing up to 66% missing values to produce a clean 920-patient cohort.
- Trained a Gradient Boosting Classifier (XGBoost-equivalent) achieving AUC 0.892 and 84.2% accuracy with 5-fold stratified cross-validation, outperforming Random Forest and Logistic Regression baselines.
- Deployed a clinical Flask dashboard with PostgreSQL persistence, SHAP-style per-patient feature contribution analysis, and population-level analytics across 4 institutional data sources.

Real-Time Web Application Personal Project

Rankr — Collaborative Real-Time Ranking App

July 2024

github.com/Chenry513/rankr

- Built a real-time polling system enabling groups to create, join, and rank shared options collaboratively.
- Implemented live updates using React, TypeScript, Socket.io, and Redis for low-latency synchronization.
- Designed backend logic to aggregate ranked votes and compute final group decisions.